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Associations of inflammatory and reproductive tract disorders postpartum with pregnancy and early pregnancy loss in dairy cows

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ABSTRACT

Our objective was to describe associations of postpartum health with pregnancy and pregnancy loss (P-LOSS) from d 19 to 40 after first postpartum artificial insemination (AI) in lactating Holstein cows. In 2 commercial dairy herds in Ontario, Canada, 468 Holstein cows were enrolled 21 ± 3 d before expected parturition when body condition score (BCS) and lameness were assessed. Serum total Ca, haptoglobin (Hp), and nonesterified fatty acids (NEFA) were measured at 2 and 6 ± 2 DIM. Blood β-hydroxybutyrate (BHB) measurement and metritis detection were done at 4, 8, 11, and 15 ± 2 DIM. Cows were examined for endometritis (ENDO; ≥ 11.5% polymorphonuclear cells in endometrial cytology) and purulent vaginal discharge (PVD) at 35 ± 3 DIM. Lameness was assessed again at 21 and 49 ± 3 DIM and BCS at 63 ± 3 DIM. First postpartum AI occurred primarily (86%) based on detection of estrus by activity monitors, on average (±SD) at 65 ± 9 DIM, and the remaining cows received timed AI at 86 ± 18 DIM. Serum progesterone (P4) was measured on d 8 and 12 after AI, and pregnancy at first AI (P/AI) was estimated by the expression of *ISG15* in peripheral blood leukocytes at d 19 after AI and by pregnancy-associated glycoprotein in serum at d 29, 33, and 40 after AI. Each metabolite (Ca, Hp, NEFA, and BHB) was categorized above or below a cut-point identified with receiver operating characteristic curve analysis associated with P/AI confirmed by ultrasound at d 33 from a larger data set. Data were analyzed using multivariable mixed logistic regression models, accounting for parity, health variables, covariates (season at calving and at AI, milk yield at first DHIA test (categorized into terciles), AI method, and DIM), and herd. The proportions of cows classified pregnant at d 19, 29, 33, and 40 after AI were 64, 54, 50, and 45%, respectively. At d 19 after AI, P/AI was less likely in cows diagnosed with ENDO (52 vs. 69%) or PVD (54 vs. 67%). At d

29, P/AI was less likely in cows with Hp ≥ 1.54 g/L at 2 DIM (38 vs. 55%) or PVD (35 vs. 56%). Both metritis and ENDO were associated with decreased P/AI at d 40 after AI. Cows diagnosed with metritis had greater risk of P-LOSS from d 19 to 29 (43 vs. 22%) or from d 33 to 40 (37 vs. 7%) than cows without metritis. From d 29 to 33, the risk of P-LOSS was greater in cows with NEFA ≥ 0.73 mM at 2 DIM (13 vs. 5%) or BCS ≤ 2.75 at 63 DIM (14 vs. 5%). The concentration of P4 on d 8 after AI was positively associated with P/AI at d 29, 33, and 40, and negatively associated with P-LOSS from d 19 to 29. Postpartum health disorders, particularly reproductive tract disease, can have detrimental impacts on early pregnancy establishment and on pregnancy maintenance from d 19 to 40 after AI.

Key words: disease, fertility, reproduction, transition health

INTRODUCTION

Poor fertility in dairy cows is an important element of economic losses to dairy producers because it extends calving interval, reduces milk productivity, and increases risk of culling (Ribeiro et al., 2012). As summarized by Fricke and Wiltbank (2022), reproductive performance decreased in the USA from 1960 to the 2000s, then phenotypic trends for P/AI improved from about 33% in 2005 to over 46% in 2020. Still, a substantial proportion of cows may lose their pregnancy either before or after the first pregnancy diagnosis at approximately 30 d after artificial insemination (AI) (Wiltbank et al., 2016), resulting in further economic losses (De Vries, 2006).

Following AI, fertilization and production of a viable embryo is expected in most cows, but assessment of pregnancy status in the early stages is difficult. There is considerable variation among studies, with 52 to 71% of lactating cows reported to have a viable embryo recovered at uterine flush by 6 to 7 d after AI (Wiltbank et al., 2016). At approximately d 18, pregnancy establishment depends on maternal recognition of the conceptus or “rescue” of the corpus luteum (CL), char-

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acterized by secretion of interferon-tau (IFN- τ) by the trophectoderm that will inhibit expression of oxytocin receptors in the endometrium and prevent endometrial luteolytic pulses of prostaglandin-F $_{2\alpha}$ (PGF $_{2\alpha}$) (Spencer and Bazer, 2004). Using the expression of an interferon-stimulated gene (*ISG15*) in peripheral blood leukocytes (PBL) as an indirect method to assess exposure to IFN- τ by d 19 after AI, pregnancy losses by 27 to 32 d after AI averaged 27% (Albaaj et al., 2023). These data suggest that substantial pregnancy loss (P-LOSS) occurs during or just after the period of maternal recognition of pregnancy and before first pregnancy diagnosis, commonly around 30 d after AI.

Besides cow biology, genetics, herd management, or environmental factors that contribute to pregnancy success (Santos et al., 2010; Wiltbank et al., 2016), postpartum health disorders are critical risk factors for reproductive losses, and can affect nearly 50% of cows (LeBlanc, 2012; Ribeiro et al., 2013). In a compilation of data from 5,500 cows, P/AI at first diagnosis at ~30 d was 51% in healthy cows, with 9% of them undergoing P-LOSS by d 60 after AI (Santos et al., 2010). Pregnancy loss was more likely in cows that had a calving problem (16%), purulent vaginal discharge (PVD; 15%), mastitis (20%), or lameness (26%). Similarly, others reported a 9% probability of P-LOSS between first pregnancy diagnosis at 30 d and second diagnosis at 60 or 65 d of gestation in healthy cows but 1.4 to 1.7 greater odds of loss in cows that experienced one or 2 reproductive tract diseases, respectively (Pinedo et al., 2020), and 3.6 times greater odds in cows that experienced multiple postpartum clinical diseases (Ribeiro et al., 2013). Although the association between postpartum health and pregnancy maintenance after pregnancy diagnosis is well established, less is known regarding the effects of health disorders on early pregnancy maintenance and P-LOSS before first pregnancy diagnosis.

Therefore, our objective was to describe associations of postpartum health with P/AI and P-LOSS from 19 to 40 d after first AI in lactating Holstein cows. Secondary objectives were to assess the relationships of serum progesterone (P4) concentrations after AI with P/AI and P-LOSS, and of serum pregnancy-associated glycoprotein (PAG) levels in cows classified as pregnant based on PAG with subsequent P-LOSS. We hypothesized that cows affected by health disorders would have reduced P/AI and be more likely to undergo early P-LOSS than unaffected cows, and that lesser serum P4 concentrations and PAG levels would be associated with subsequent P-LOSS.

MATERIALS AND METHODS

This study was approved by the Animal Care Committee of the University of Guelph (AUP #4385) and the manuscript is reported using the STROBE-Vet checklist (Strengthening the Reporting of Observational Studies in Epidemiology – Veterinary Extension) (O'Connor et al., 2016). The data reported here are a subset from a larger observational study on factors associated with detection of estrus for first AI and pregnancy outcomes (Bruinje et al., 2023).

Description of herds

Two commercial dairy herds in southwestern Ontario were selected for enrollment due to their proximity to the University of Guelph, consistent record keeping for reproductive management decisions, and willingness to participate in the study. Herds milked approximately 450 Holstein cows each using either a parallel parlor system (3 milking/d; herd A) or an automated milking system (3.2 milking/d; herd B), with average lactation milk yield (305 d) of 11,558 kg for primiparous and 14,256 kg for multiparous cows in herd A, and 9,931 kg for primiparous and 12,495 kg for multiparous cows in herd B. Lactating cows were housed in freestall barns in both herds and feed was delivered 2 or 3 times daily in herd A and once daily in herd B.

Overall reproductive performance metrics during the study, retrieved from DairyComp305 (Valley Ag Software) in herd A and B, respectively, were 25 and 23% 21-d pregnancy rate, 62 and 63% 21-d insemination rate, and 44 and 39% P/AI. Management for first postpartum AI was primarily based on detection of estrus by automated activity monitors with pedometers (AfiAct, Afimilk, Kibutz Afikim) starting at 55 DIM (herd A) or with collars (SmartTag Neck, Nedap Livestock Management) starting at 50 DIM (herd B). Cows were inseminated within 12 h (herd A) or 24 h (herd B) following an estrus alert generated by the activity system. In herd A, cows that were not detected in estrus received a PGF $_{2\alpha}$ (500 μ g cloprostenol sodium; Bioestrovect, Vetoquinol or 25 mg dinoprost tromethamine; Lutalyse, Zoetis) injection at 73 DIM and, if not detected in estrus by 87 DIM, received a timed AI protocol [Ovsynch: GnRH (100 μ g gonadorelin acetate; Fertiline, Vetoquinol) – 7 d – PGF $_{2\alpha}$ – 56 h – GnRH – 12 h – AI]. Cows without a CL at ultrasound examination at the start of Ovsynch received an intravaginal progesterone device (1.38 g progesterone; CIDR, Zoetis) from d 0 to 7, and multiparous cows received a second PGF $_{2\alpha}$ at d 8. In herd B, primiparous cows not detected in estrus by 78 DIM were administered an Ovsynch protocol if they had a CL, or received a GnRH (Fertiline, Veto-

quinol) injection followed by Ovsynch 7 d later if did not have a CL. Multiparous cows not detected in estrus and with a CL received an Ovsynch protocol at ~65 DIM, or were administered GnRH injection followed by an Ovsynch 7 d later if did not have a CL. During the study, 86% of cows were bred after detection of estrus on average ($\pm$ SD) at 65 ± 9 DIM, with the remainder bred following synchronization at 86 ± 18 DIM.

Assessment of postpartum health

Herds were visited twice weekly between May 2019 and February 2021 by the research team. A total of 1,646 Holstein cows were initially enrolled, as part of a larger study, at 257 ± 3 d of gestation, from which a subsample of 469 cows [150 nulliparous (79 in herd A and 71 in herd B) and 319 parous (185 in herd A and 134 in herd B)] were randomly selected for the present study for data collection after first AI. Following first postpartum AI, the present subsample of cows ($n = 469$) was randomly selected for this study based on the day of the week that first AI occurred, such that sampling on farm visit days occurred ± 1 d from the target sampling day after AI, as described below. At initial enrollment, prepartum cows were assessed for body condition score [BCS (0 to 5 scale; Edmonson et al., 1989)] and locomotion score (0 to 5 scale; Sprecher et al., 1997). At 2 and 6 ± 2 DIM, blood samples were taken from the middle coccygeal vein or artery in 10-mL tubes without anticoagulant (Vacutainer Precision Glide, Becton Dickinson), centrifuged at $1,500 \times g$ for 15 min within 4 h, and stored at -20°C until analyzed with an automated biochemistry analyzer (Cobas 6000 c 501, Roche Diagnostics) to measure serum concentrations of total calcium (Ca), haptoglobin (Hp), and NEFA at the Animal Health Laboratory (University of Guelph). Total Ca was measured using the Cobas Calcium Gen.2 kit (Roche Diagnostics), NEFA using the Randox Manual Rx Monza kit (Randox Laboratories), and Hp according to a previously described protocol based on methemoglobin reduction (Makimura and Suzuki, 1982; Skinner et al., 1991). The lower limit of quantification (LLOQ) was 0.63 mM for total Ca, 0.05 mM for NEFA, and 0.02 g/L for Hp, and the intraassay coefficients of variation were 1.8%, 6.7%, and 7.7%, respectively. Blood β -hydroxybutyrate (BHB) was measured at 4, 8, 11, and 15 ± 2 DIM using a hand-held device (Precision Xtra Meter and B-ketone test strips, Abbott Laboratories).

Uterine health was evaluated using an intravaginal device (Metricheck, Simcro) at 4, 8, 11, and 15 ± 2 DIM, and a fetid, watery, and brownish or reddish vaginal discharge was classified as metritis, regardless of rectal temperature. At 35 ± 3 DIM, vaginal con-

tent was sampled using the Metricheck and cows with mucopurulent or purulent discharge [score 2 or 3 as in Sheldon et al. (2006)] were classified as having PVD. At the same time, endometrial cytology was obtained using a cytobrush at the dorsal aspect of the uterine body and slides were prepared and stained using a commercial kit (Hema 3 Stat Pack; Fisher Scientific). The proportion of polymorphonuclear cells (PMN) among endometrial cells was evaluated over 300 cell structures with an optical microscope at $400 \times$ magnification.

Serum P4 concentrations were measured at 21, 35, 49, and 63 ± 3 DIM with a commercial ELISA (Ovucheck Plasma, Biovet) validated for use in serum (Broes and LeBlanc, 2014). The threshold to determine cyclicity was 1.0 ng/mL and time to onset of cyclicity was estimated by the first sample with $P4 \geq 1.0$ ng/mL. The LLOQ was 0.89 ng/mL and inter- and intraassay coefficients of variation were 6.7% and 8.9%, respectively. Lameness (locomotion score ≥ 3) was assessed again at 21 and 49 ± 3 DIM, and BCS at 63 ± 3 DIM. The median prepartum BCS was 3.50 and prepartum BCS was categorized as ≤ 3.50 or ≥ 3.75 , BCS at 63 DIM was categorized as ≤ 2.75 or ≥ 3.00 , and the difference between pre- and postpartum BCS (median = 0.50) was categorized as no loss, 0.25-point loss, or ≥ 0.50 -point loss. Additional information on calving ease, retained placenta, milk fever, displaced abomasum (DA), and clinical mastitis (with a recorded treatment) by 30 DIM were obtained from farm records.

All multiparous cows received oral Ca bolus supplementation immediately after calving, and cows were treated for elevated BHB (≥ 1.2 mM) and for clinical diseases according to each farm's protocols as previously described (Bruinje et al., 2023). Milk yield data at the first DHIA test at $38 (\pm 18)$ DIM were obtained from Lactanet, and categorized as lowest, middle, or highest tercile of production (kg/d) for each parity group in each herd. In the lowest, middle, and highest tercile groups in herd A, primiparous cows averaged 31.4, 38.0, 44.3 kg/d and multiparous cows averaged 47.0, and 57.6, and 66.7 kg/d, respectively. In herd B, primiparous cows averaged 23.3, 31.2, and 38.2 kg/d and multiparous cows averaged 34.2, 45.4, and 54.7 kg/d respectively.

Determination of pregnancy status

Blood samples were taken at d 8 and 12 ± 1 after first AI for measurement of serum P4 in a subset of 403 cows, when cows were expected to be in early- and mid-di estrus with intermediate and high P4 concentrations (Molina-Coto et al., 2020). At d 19 ± 1 , blood samples were obtained from 398 cows with tubes containing 6 mL RNA stabilizing reagent (Tempus Blood RNA

Tube, Applied Biosystems), kept on ice, and stored at -20°C within 4 d of collection until processed for isolation of blood leukocytes and determination of *ISG15* expression. This gene was chosen because it is among the interferon-stimulated genes shown to have a distinguishable heightened expression in PBL on d 18 to 20 of pregnancy compared with nonpregnant cows (Gifford et al., 2007). The RNA extraction was performed with the Tempus Spin RNA Isolation Kit (Applied Biosystems) as per the manufacturer's instructions, with an extra DNA removal step (RapidOut DNA Removal Kit, Thermo Scientific). An average of 82 (± 38) ng/ μL of RNA were isolated with relative concentration of nucleic acid to purified proteins (A260:A280) of 2:1. Quantitative real-time PCR were done at the Genomics Facility (Advanced Analysis Centre, University of Guelph) using a QuantStudio 7 Pro Real Time PCR system (Thermo Fisher Scientific Inc.). Briefly, 20- μL reactions comprised 1 μg RNA and 10 μL 2 $\times$ SsoAdvanced Universal Inhibitor-Tolerant SYBR Green Supermix (Bio-Rad), 0.8 μL of PCR forward and reverse primer mix at 5 μM (final primer concentration at 200 nM), 4.2 μL of water, and 5 μL of diluted cDNA. The cDNA was obtained from 1 μg RNA using a High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems). The reactions were run in duplicate and included a polymerase activation step for 3 min at 98°C , followed by 40 cycles of a 2-step qPCR (10 s of 98°C denaturation, 30 s of 60°C combined annealing/extension). The expression of *ISG15* (target gene) and of 2 reference genes [β -actin (*ACTB*) and ribosomal protein L19 (*RPL19*)] were analyzed (Table 1). The difference between the threshold cycle (C_T) of the *ISG15* and the geometric mean of the reference genes (Vandesompele et al., 2002) was used to obtain the *ISG15* ΔC_T values.

Serum PAG was assessed at d 29, 33, and 40 (± 1) after AI using a commercial ELISA (Bovine Pregnancy Test Kit, IDEXX Laboratories Inc., Westbrook, ME). Sampling days for PAG was based on a previous study reporting plasma pregnancy-specific protein B concentrations increasing in pregnant cows by d 28 after AI, followed by a peak between d 31 to 35, and a subtle decrease by d 42 in cows that subsequently lost the

pregnancy compared with cows that maintained the pregnancy (Ribeiro et al., 2014). Around these time points, cows randomly selected to be enrolled in our sampling scheme after AI, as described above, would have samples taken at d 29, 33, and 40 ± 1 . The difference between the adjusted optical density (OD) of the sample (OD at 635 nm minus OD at 450 nm) and the adjusted OD of the negative control was calculated, and the inter- and intra-assay coefficients of variation of this arbitrary unit were 1.7% and 0.9%, respectively. Samples with a ratio ≥ 0.3 were classified as positive for pregnancy according to the manufacturer's instructions, and levels of PAG were interpreted as the ratio between sample and positive control values (sample-to-positive ratio; S:P). The accuracy of diagnosing pregnancy at similar time points based on plasma PAG was $>91\%$ with a different assay (Sinedino et al., 2014), and the positive predictive value was 100% at d 32 and 39 after AI with the same assay used here (Ricci et al., 2015). Pregnancy status based on PAG at d 29 was used as the reference to identify the optimal cut-point of *ISG15* ΔC_T to predict P/AI at d 19 with the maximum combined sensitivity and specificity (Youden's index) using ROC curve analysis. Cows with *ISG15* ΔC_T value below the cut-point, as well as those above the cut-point but classified as pregnant by PAG at d 29 (false negatives by *ISG15*) were categorized as pregnant at d 19.

Statistical analysis

Data were analyzed using SAS (Studio 3.8 Edition; version 9.04; SAS Institute Inc.) and graphs were generated using GraphPad Prism (version 8.0.0; GraphPad Software). In a study with a smaller sample size (Molina-Coto et al., 2020), the estimated proportion of cows classified pregnant at d 20 based *ISG15* but not pregnant at d 32 based on US was 23 vs. 9% in cows with vs. without endometritis. It would require at least 107 cows per group to detect that difference with 80% power and 95% confidence, or 535 cows in total for a health disorder affecting 20% of cows. Nonetheless, for the present study, the sample size was based on the

Table 1. Primer sequences for the target gene (*ISG15*) and reference genes (*ACTB* and *RPL19*) for quantification of mRNA in blood leukocytes 19 d after first insemination

Gene	NCBI sequence	Primer sequence
Interferon-stimulated gene 15 (<i>ISG15</i>)	NM_174366	Forward: 5'-GGTATGAGCTGAAGCAGTT-3' Reverse: 5'-ACCTCCCTGCTGTCAAGGT-3'
β -actin (<i>ACTB</i>)	AY141970	Forward: 5'-CTGGACTTCGAGCAGGAGAT-3' Reverse: 5'-GGATGTCGACGTCACACTTC-3'
Ribosomal protein L19 (<i>RPL19</i>)	NM_001040516	Forward: 5'-ATCGATCGCCACATGTATCA-3' Reverse: 5'-GCGTGCTTCCTTGGTCTTAG-3'

availability of resources and was a subset from a larger study (Bruinje et al., 2023).

Descriptive statistics were obtained using the MEANS and FREQ procedures in SAS. Continuous variables of postpartum health were categorized based on cut-points obtained in ROC curve analysis (LOGISTIC procedure in SAS) associated with P/AI confirmed by ultrasound at d 33 in the larger study ($n = 1,454$; Bruinje et al., 2023) that included data from the same cows. For total Ca, ROC curve analysis was first tested for each parity group, but because it was less discriminatory (smaller area under the curve) than when combining parity groups, a common cut-point was used for all cows. Cut-points at 2 and 6 ± 2 DIM for total Ca were 2.09 and 2.28 mM, 0.73 and 0.63 mM for NEFA, and 1.54 and 0.62 g/L for Hp, respectively. Cut-points of BHB were 0.8 mM at 4 ± 2 DIM, 0.9 mM at 8 ± 2 DIM, and 0.7 mM at both 11 and 15 ± 2 DIM, respectively, and for endometritis (ENDO) was 11.5% PMN.

Outcomes of P/AI at d 19 and 40 and of P-LOSS were analyzed in mixed logistic regression models using the GLIMMIX procedure in SAS. Predictor variables evaluated were parity, metabolic events (hypocalcemia, elevated NEFA, elevated Hp, hyperketonemia), prepartum BCS, BCS change, BCS at 63 DIM, calving difficulty, RP, metritis, PVD, ENDO, DA, mastitis, lameness, interval to onset of cyclicity (by 49 or 63 ± 3 DIM), and milk yield, in addition to the covariates of season at AI, method for first AI (detected estrus vs. timed AI), and DIM at first AI. Concentrations of P4 at d 8 and 12 after AI and levels of PAG at d 29 (in models of P/AI and P-LOSS beyond d 29) were tested in separate models as they were considered potential intervening variables.

For each outcome, predictors or covariates were analyzed in univariable logistic regression models, and those with $P \leq 0.20$ were selected for the multivariable model that included herd as a random effect. Collinearity among predictors was checked using variance inflation factors, and the final model was selected by backward manual stepwise elimination maintaining all variables with $P \leq 0.10$. The variance inflation factor was always <1.6 , which indicated no issues with collinearity. Removed variables were individually reentered in the final models and retained as confounders if their inclusion changed the regression coefficient of the significant variables $>20\%$. In addition, separate models that included disease variables combined either as clinical disease (any of milk fever, DA, mastitis, lameness, RP, metritis, or PVD vs. none), or type of disease [uterine (UTD; any of RP, metritis, ENDO, or PVD), non-uterine clinical disease (NUTD; any of milk fever, DA, lameness, or mastitis), both, or none] were also analyzed. For pregnancy outcomes, similar mul-

tivariable models were built but including only cows that had P4 concentration ≥ 1.0 ng/mL at d 8 or 12 among the subset that had these P4 measurements. As described below, this excluded a small number of cows that were unlikely to be pregnant.

RESULTS

In total, 383 cows were evaluated for P/AI at d 19 ± 1 based on *ISG15*, and 469 for P/AI at d 29, 33, and 40 ± 1 based on PAG. Cows were excluded from the final models if they had missing measurements on the predictor variables ($n = 3$ for P/AI at d 19, $n = 4$ for P/AI at d 40, and $n = 1$ for P-LOSS from d 29 to 33). Among 403 cows with serum P4 measurement after AI, 6 and 3%, had $P4 < 1.0$ ng/mL on d 8 and 12 after AI, respectively, suggesting that these cows were not inseminated at the correct time or failed to ovulate after AI. The prevalence of clinical health disorders in the larger data set was reported previously (Bruinje et al., 2023). The incidence or prevalence of health variables classified based on blood markers of examined clinically among cows included in the analysis are presented in Table 2. Fifty-eight percent (271/469) of cows were clinically healthy and 29% (135/469) and 13% (63/469) were diagnosed with a single or with >1 clinical disorder, respectively. The incidence risk of UTD and NUTD was 37% (175/469) and 20% (95/469), respectively, and 8% (40/469) of cows were diagnosed with both UTD and NUTD.

Based on ROC curve analysis, the cut-point of *ISG15* to predict P/AI at d 19, using pregnancy based on PAG at d 29 as the reference (outcome), was $7.24 \Delta C_T$. This cut-point resulted in 88% sensitivity (12% false negatives), 70% specificity, 74% positive predictive value, and 86% negative predictive value to predict pregnancy, with an area under the ROC curve of 82%. The total proportions of cows classified pregnant at d 19 (based on *ISG15*), and on d 29, 33, and 40 (based on PAG) were 64% (245/382), 54% (252/469), 50% (236/469), and 45% (209/469), respectively.

Among cows with $P4 \geq 1.0$ ng/mL on d 8 or 12 after AI, variables associated with reduced P/AI at d 19 in the multivariable model were parity (multiparous; $P = 0.02$), ENDO ($P = 0.02$), and PVD ($P = 0.07$) (Table 3). Variables associated with reduced P/AI at d 29 were parity (multiparous; $P < 0.01$), elevated Hp (≥ 1.54 g/L) at 2 DIM ($P = 0.02$), and PVD ($P < 0.01$) (Table 4). Similarly, risk factors for reduced P/AI at d 33 were parity (multiparous; $P < 0.01$), elevated Hp (≥ 1.54 g/L) at 2 DIM ($P = 0.05$), and PVD ($P < 0.01$) (Table 5), while risk factors for reduced P/AI at d 40 were parity (multiparous; $P < 0.01$), metritis ($P < 0.01$), and ENDO ($P = 0.05$) (Table 6). From d 19

Table 2. Incidence of transition health disorders categorized based on blood markers or examined clinically in Holstein cows enrolled from two commercial dairy herds in Ontario, Canada

Variable	Definition ¹	Prevalence or incidence (%); Affected / Total (n/n)
Hypocalcemia ²		
2 ± 2 DIM	≤2.09 mM	55% (258/467)
6 ± 2 DIM	≤2.28 mM	60% (274/460)
Elevated Hp ²		
2 ± 2 DIM	≥1.54 g/L	18% (82/467)
6 ± 2 DIM	≥0.62 g/L	39% (178/460)
Elevated NEFA ²		
2 ± 2 DIM	≥0.73 mM	23% (105/467)
6 ± 2 DIM	≥0.63 mM	32% (145/460)
Hyperketonemia ²		
4 ± 2 DIM	≥0.8 mM	44% (205/466)
8 ± 2 DIM	≥0.9 mM	23% (108/462)
11 ± 2 DIM	≥0.7 mM	54% (250/467)
15 ± 2 DIM	≥0.7 mM	57% (263/462)
Retained placenta ³		8% (35/469)
Metritis ³		14% (65/469)
Endometritis ³	≥11.5% PMN	23% (105/465)
PVD ³		19% (90/469)
Lameness ⁴		17% (79/469)
Displaced abomasum		3% (13/469)
Uterine disease ⁵		37% (175/469)
Clinical disease ⁶		
One		29% (135/469)
Multiple		13% (63/469)

¹Thresholds were selected based on the maximum combined sensitivity (Se) and specificity (Sp) values to predict pregnancy at d 33 after first insemination in a larger data set (n = 1,454 cows).

²Based on total Ca, haptoglobin (Hp), and nonesterified fatty acids (NEFA) concentrations measured in serum, and blood β-hydroxybutyrate.

³Retained placenta = failure to expel fetal membranes by 24 h after calving; Metritis = detection of fetid and colored (brownish/red-dish) discharge at any of four examinations (4, 8, 11, and 15 ± 2 DIM); Endometritis = based on percentage of polymorphonuclear cells (PMN) assessed in endometrial cytology at 35 ± 3 DIM; PVD = mucopurulent or purulent vaginal discharge at 35 ± 3 DIM.

⁴Lameness: locomotion score ≥3 (1 to 5 scale) at 21 or 49 ± 3 DIM.

⁵Diagnosed with at least one of retained placenta, metritis, endometritis, or PVD.

⁶Diagnosed with at least one of milk fever, retained placenta, metritis, PVD, displaced abomasum by 30 DIM, mastitis by 30 DIM, or lameness.

through 40 after AI, P/AI was lesser in cows diagnosed with UTD only (excluding those with a NUTD) compared with clinically healthy cows (Figure 1), or cows diagnosed with >1 clinical disease (Figure 2). There was no association between NUTD and P/AI.

Each 1.0 ng/mL increase in serum P4 concentration on d 8 after AI was associated with 18% greater odds (AOR: 1.18; 1.04 to 1.34; $P = 0.01$) of P/AI at d 29, 22% greater odds (AOR: 1.22; 1.07 to 1.38; $P < 0.01$) of P/AI at d 33, and 12% greater odds (AOR: 1.12; 0.99 to 1.27; $P = 0.07$) of P/AI at d 40. The associations of P4 concentration on d 8 and the model-predicted probability of P/AI on d 29 and 33 were similar, so only the

linear associations for P/AI on d 29 and 40, accounting for covariates, are presented in Figure 3.

Risk factors for P-LOSS from d 19 to 29, 29 to 33, and 33 to 40 are presented in Table 7. The odds of P-LOSS from d 19 to 29 were nearly 3 times greater in cows affected by metritis or by >1 clinical disease compared with unaffected cows. From d 29 to 33, the odds of P-LOSS were greater in cows that had NEFA ≥0.73 mM at 2 ± 2 DIM or BCS ≤2.75 at 63 DIM. From d 33 to 40 after AI, P-LOSS was more likely in cows diagnosed with metritis, DA, or with >1 clinical disease, in a multivariable model that controlled for season of AI. Overall, from d 19 to 40, 36% (88/245) of cows had P-LOSS, which was 4 to 5 times more likely in cows diagnosed with metritis, DA, or >1 clinical disease (Table 8).

Each 1 ng/mL increase in serum P4 concentration on d 8 after AI was associated with 30% decreased odds (AOR: 0.70; 0.56 to 0.88; $P < 0.01$) of P-LOSS from d 19 to 29 and 18% decreased odds (AOR: 0.82; 0.68 to 0.99; $P = 0.04$) of P-LOSS from d 19 to 40 (Figure 4). For each 1.0 unit increase in serum PAG S:P, the odds of P-LOSS from d 29 to 33 decreased by 75% (AOR: 0.25; 0.14 to 0.44; $P < 0.001$) and the odds of P-LOSS from d 33 to 40 decreased by 58% (AOR: 0.42; 0.27 to 0.65; $P < 0.01$) (Figure 5). There were no associations of other health variables (hypocalcemia, hyperketonemia, prepartum BCS, BCS change, difficult calving, RP, mastitis, delayed cyclicity by 49 or 63 DIM) or covariates (milk yield, method for first AI, DIM at first AI) with P/AI or P-LOSS in the multivariable models.

DISCUSSION

Uterine disease was the major risk factor for decreased odds of early pregnancy and increased risk of early P-LOSS. Specifically, P/AI at d 19 was reduced in cows diagnosed with ENDO or with any UTD, and P/AI at d 33 and 40 was reduced in cows diagnosed with metritis, ENDO, > 1 clinical disease, or with any UTD. Cows diagnosed with metritis or >1 clinical disease were more likely to have P-LOSS from d 19 to 29 and from d 33 to 40, while P-LOSS from d 29 to 33 was more likely in cows that had NEFA ≥0.73 mM at 2 ± 2 DIM or BCS ≤2.75 at 63 DIM.

The proportions of cows classified pregnant from d 19 to 40 were comparable to other reports. As early as 6 to 7 d after AI, between 52 and 71% of lactating dairy cows had a viable embryo in uterine flushing under thermoneutral conditions (as summarized in Wiltbank et al., 2016). Here, 64% of cows were estimated to be pregnant at d 19 after AI based on *ISG15*, and 54%, 50%, and 45% were diagnosed pregnant at d 29, 33, and 40 based on PAG, respectively. In a summary of 4

Table 3. Final logistic regression models of variables associated with presumed pregnancy at d 19 after first artificial insemination (AI) based on expression of *ISG15* in blood leukocytes in Holstein cows from two herds

Variable	n	LSM ¹	AOR ¹	95% CI ¹	P-value
All cows (n = 379)					
Parity					
Primiparous	117	71.9%	Reference	—	—
Multiparous	262	60.3%	0.59	0.36 to 0.97	0.04
Endometritis²					
Unaffected	291	68.7%	Reference	—	—
Affected	88	48.1%	0.42	0.26 to 0.69	<0.01
Cows with P4 ≥ 1.0 ng/mL after AI (n = 348)³					
Parity					
Primiparous	101	74.9%	Reference	—	—
Multiparous	247	61.8%	0.52	0.30 to 0.89	0.02
Endometritis					
Unaffected	265	68.6%	Reference	—	—
Affected	83	52.4%	0.5	0.29 to 0.88	0.02
PVD⁴					
Unaffected	281	67.4%	Reference	—	—
Affected	67	54.2%	0.57	0.31 to 1.04	0.07

¹Least squares means (LSM) of predicted probability of pregnancy, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

²≥ 11.5% polymorphonuclear cells in endometrial cytology performed at 35 ± 3 DIM.

³Includes only cows that had serum progesterone (P4) concentration ≥1.0 ng/mL at d 8 or 12 after AI, among 359 cows with serum P4 measured.

⁴Mucopurulent or purulent vaginal discharge (PVD) examined at 35 ± 3 DIM.

studies and over 700 cows, 70% P/AI was reported at d 18 or 19 based on *ISG15*, with 20% P-LOSS from d 19 to 27 and 32% P-LOSS from d 19 to 32 (Wiltbank et al., 2016). The apparent accuracy of *ISG15* for P/AI in

a study with 78 cows (Han et al., 2006), with 81% sensitivity and 75% specificity, was similar to our results. Using cows not bred as negative controls, Wijma et al. (2016) reported 74% sensitivity and 92% specific-

Table 4. Final logistic regression models of variables associated with pregnancy at d 29 after first artificial insemination (AI) based on serum pregnancy-associated glycoprotein in Holstein cows from two herds

Variable	n	LSM ¹	AOR ¹	95% CI ¹	P-value
All cows (n = 469)					
Parity					
Primiparous	150	63.8%	Reference	—	—
Multiparous	319	47.9%	0.52	0.35 to 0.79	<0.01
Metritis²					
Unaffected	404	56.0%	Reference	—	—
Affected	65	36.3%	0.45	0.25 to 0.81	<0.01
PVD²					
Unaffected	379	56.0%	Reference	—	—
Affected	90	41.9%	0.57	0.34 to 0.94	0.03
Cows with P4 ≥ 1.0 ng/mL after AI (n = 382)³					
Parity					
Primiparous	110	64.4%	Reference	—	—
Multiparous	272	47.0%	0.49	0.30 to 0.80	<0.01
Hp at 2 DIM⁴					
<1.54 g/L	312	54.9%	Reference	—	—
≥1.54 g/L	70	38.4%	0.51	0.29 to 0.91	0.02
PVD²					
Unaffected	310	55.9%	Reference	—	—
Affected	72	34.6%	0.42	0.24 to 0.74	<0.01

¹Least squares means (LSM) of predicted probability, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

²Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 ± 2 DIM; PVD = mucopurulent or purulent vaginal discharge at 35 ± 3 DIM.

³Includes only cows that had serum progesterone (P4) concentration ≥1.0 ng/mL at d 8 or 12 after AI, among 403 cows with serum P4 measured.

⁴Serum haptoglobin (Hp) concentration measured at 2 ± 2 DIM.

Table 5. Final logistic regression models of variables associated with pregnancy (P/AI) at d 33 after first artificial insemination (AI) based on serum pregnancy-associated glycoprotein in Holstein cows from two herds

Variable	n	LSM ¹	AOR ¹	95% CI ¹	P-value
All cows (n = 465)					
Parity					
Primiparous	148	59.3%	Reference	—	—
Multiparous	317	44.7%	0.55	0.37 to 0.84	<0.01
Metritis²					
Unaffected	401	52.6%	Reference	—	—
Affected	64	31.0%	0.40	0.22 to 0.74	<0.01
Endometritis³					
Unaffected	360	52.5%	Reference	—	—
Affected	105	39.3%	0.59	0.36 to 0.95	0.03
Cows with P4 $\geq$ 1.0 ng/mL after AI (n = 382)⁴					
Parity					
Primiparous	110	60.4%	Reference	—	—
Multiparous	272	43.4%	0.5	0.31 to 0.81	<0.01
Hp at 2 DIM⁵					
<1.54 g/L	312	50.7%	Reference	—	—
$\geq$ 1.54 g/L	70	36.4%	0.55	0.31 to 0.99	0.05
PVD³					
Unaffected	310	52.1%	Reference	—	—
Affected	72	31.5%	0.42	0.24 to 0.75	<0.01

¹Least squares means (LSM) of predicted probability, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

²Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 $\pm$ 2 DIM).

³Endometritis was defined as $\geq$ 11.5% polymorphonuclear cells in endometrial cytology performed at 35 $\pm$ 3 DIM; PVD = mucopurulent or purulent vaginal discharge at 35 $\pm$ 3 DIM.

⁴Includes only cows that had serum progesterone (P4) concentration $\geq$ 1.0 ng/mL at d 8 or 12 after AI, among 396 cows with serum P4 measured.

⁵Serum haptoglobin (Hp) concentration measured at 2 $\pm$ 2 DIM.

Table 6. Final logistic regression models of variables associated with pregnancy (P/AI) at d 40 after first artificial insemination (AI) based on serum pregnancy-associated glycoprotein in Holstein cows from two herds

Variable	n	LSM ¹	AOR ¹	95% CI ¹	P-value
All cows (n = 465)					
Parity					
Primiparous	148	54.0%	Reference	—	—
Multiparous	317	37.3%	0.51	0.33 to 0.77	<0.01
Metritis²					
Unaffected	401	47.2%	Reference	—	—
Affected	64	19.1%	0.26	0.13 to 0.53	<0.01
Endometritis³					
Unaffected	360	46.0%	Reference	—	—
Affected	105	31.3%	0.53	0.32 to 0.88	0.02
Cows with P4 $\geq$ 1.0 ng/mL after AI (n = 380)⁴					
Parity					
Primiparous	108	53.1%	Reference	—	—
Multiparous	272	36.3%	0.5	0.31 to 0.81	<0.01
Metritis²					
Unaffected	324	45.2%	Reference	—	—
Affected	56	18.3%	0.27	0.13 to 0.58	<0.01
Endometritis³					
Unaffected	288	43.9%	Reference	—	—
Affected	92	31.2%	0.58	0.33 to 1.01	0.05

¹Least squares means (LSM) of predicted probability, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

²Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 $\pm$ 2 DIM).

³Endometritis was defined as $\geq$ 11.5% polymorphonuclear cells in endometrial cytology performed at 35 $\pm$ 3 DIM.

⁴Includes only cows that had serum progesterone (P4) concentration $\geq$ 1.0 ng/mL at d 8 or 12 after AI, among 394 cows with serum P4 measured.

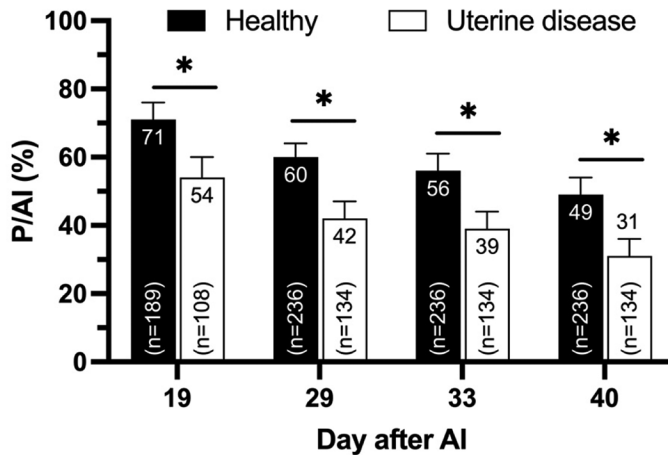


Figure 1. Predicted probability of pregnancy at first artificial insemination (P/AI) in lactating Holstein cows diagnosed with postpartum uterine disease (any of retained placenta, metritis, purulent vaginal discharge, or endometritis) compared with healthy cows not diagnosed with any uterine disease or non-uterine disease ($*P \leq 0.05$) evaluated in a multivariable model controlling for parity and body condition score as covariates, and herd as a random effect. Data exclude cows diagnosed with a non-uterine clinical disease ($n = 85$ at d 19; $n = 99$ at d 29, 33, and 40). Pregnancy status was estimated at d 19 after AI based on expression of *ISG15* in blood leukocytes and at d 29, 33, and 40 based on serum pregnancy-associated glycoprotein.

ity using *ISG15* to detect the presence of an embryo. Han et al. (2006) observed 50% P/AI at d 19 based on *ISG15* and 33% P-LOSS from d 19 to 33. Similarly, we observed 24% P-LOSS from d 19 to 29 and 28%

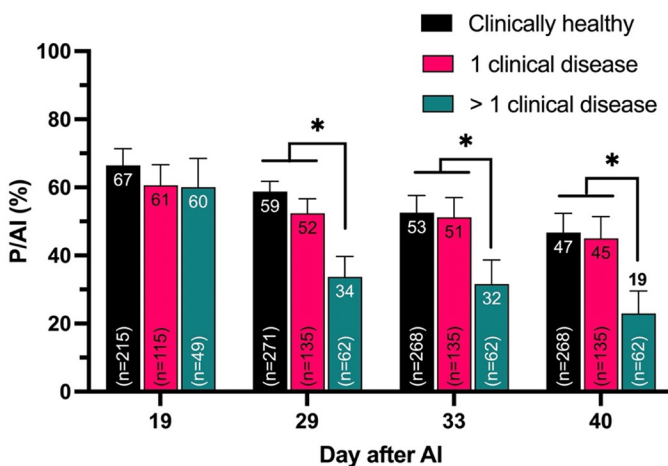


Figure 2. Predicted probability of pregnancy at first artificial insemination (P/AI) in lactating Holstein cows diagnosed with none (clinically healthy), one, or multiple (>1) postpartum clinical diseases (any of retained placenta, milk fever, displaced abomasum, lameness, mastitis, metritis, or purulent vaginal discharge) ($*P \leq 0.05$) evaluated in a multivariable model controlling for parity and subclinical endometritis as covariates, and herd as a random effect. Pregnancy status was estimated at d 19 after AI based on expression of *ISG15* in blood leukocytes and at d 29, 33, and 40 based on serum pregnancy-associated glycoprotein.

P-LOSS from d 19 to 33. Our findings support that: 1) most cows likely have a viable embryo at d 19 after AI; 2) at least a quarter of cows undergo P-LOSS before first pregnancy diagnosis at around 30 d of gestation; 3) P-LOSS before first pregnancy diagnosis is substantially worsened by uterine disease approximately 2 mo earlier.

There is limited literature reporting risk factors for early P/AI and P-LOSS during the period of maternal recognition of pregnancy, which occurs between d 16 and 25 after breeding (Wiltbank et al., 2016). On d 5 or 6 after AI, a lower proportion of live embryos were recovered in cows that had UTD (~50 vs. 70%) or NUTD (~60 vs. 70%) compared with healthy cows (Ribeiro et al., 2016). Ribeiro et al. (2016) also reported smaller conceptuses at d 15 and lesser concentration of IFN- τ in the uterine flushing of cows that previously had UTD or NUTD than in the uterine flushing of cows that had not experienced clinical diseases, although the proportions of cows with IFN- τ detected in the uterine flush at d 15 were similar. With our findings of decreased P/AI at d 19 in cows diagnosed with ENDO or with >1 clinical disease, we speculate that embryos of cows affected by uterine disease might have compromised development and less IFN- τ secretion than embryos from healthy cows. We also observed lesser P/AI at d 40 and greater risk of P-LOSS from d 19 to 29 and from d 33 to 40 in cows diagnosed with metritis. In a previous study, cows diagnosed with metritis had indications of compromised endometrial glandular development one month postpartum and during the breeding period compared with clinically healthy cows (Sellmer Ramos et al., 2023). Impairment of endometrial function or of early conceptus signaling during maternal recognition of pregnancy could explain the decreased P/AI and increased risk of early P-LOSS observed here in cows that were diagnosed with metritis.

Uterine disease or the occurrence of multiple clinical diseases were risk factors for decreased P/AI from d 29 to 40 after AI. Such associations have been similarly observed in larger studies evaluating pregnancy at first pregnancy diagnosis (Santos et al., 2010; Dubuc et al., 2011; Ribeiro et al., 2016), including the larger study from which the present data are drawn (Bruinje et al., 2023). Interestingly, in cows with serum P4 ≥ 1.0 ng/mL after AI, elevated Hp at 2 DIM was associated with reduced P/AI at d 29 and 33. This suggests that excessive systemic inflammation shortly after parturition might have long-term consequences for pregnancy establishment, even when accounting for other clinical diseases.

Lesser serum P4 concentration during early diestrus after AI was associated with lower P/AI from d 29 to 40, and greater P-LOSS from d 19 to 29 and from d

19 to 40. Suboptimal circulating P4 during the early diestrus phase following fertilization altered expression of genes in the endometrium and worsened embryonic development (Forde et al., 2009, 2011). In a larger observational study, lower milk P4 concentration during early diestrus after AI was associated with reduced P/AI assessed at d 50 after AI (Bruinje et al., 2019). It is also possible that some cows were bred at false positive estrus alerts or were asynchronous at the time of AI and ovulated later, resulting in lower P4 at d 8 after AI and reduced probability of pregnancy. This could have occurred if the estrus detection by AAM and AI occurred slightly later than the optimal timing of AI, or if cows had incomplete luteolysis or delayed luteinization during the timed AI protocol. We report models evaluating associations between serum P4 and pregnancy

outcomes including only cows with serum P4 ≥ 1.0 ng/mL on d 8 or 12 after AI. Our findings suggest that possibly because of impaired uterine environment and embryonic development, cows presumably pregnant at d 19 were more likely to undergo P-LOSS by d 29.

Few studies looked at risk factors for P-LOSS before d 40 of gestation. Molina-Coto et al. (2020) reported greater P-LOSS from d 22 (based on *ISG15*) to d 45, or from d 32 to d 45 in cows diagnosed with a uterine disease compared with healthy cows. A study with approximately 1,400 lactating cows observed greater P-LOSS from d 31 to 45 in cows diagnosed with mastitis, but other clinical diseases were not evaluated (Chebel et al., 2004). In the present study, cows with elevated NEFA at 2 DIM or with low BCS at 63 DIM were more likely to undergo P-LOSS from d 33 to 40, although the

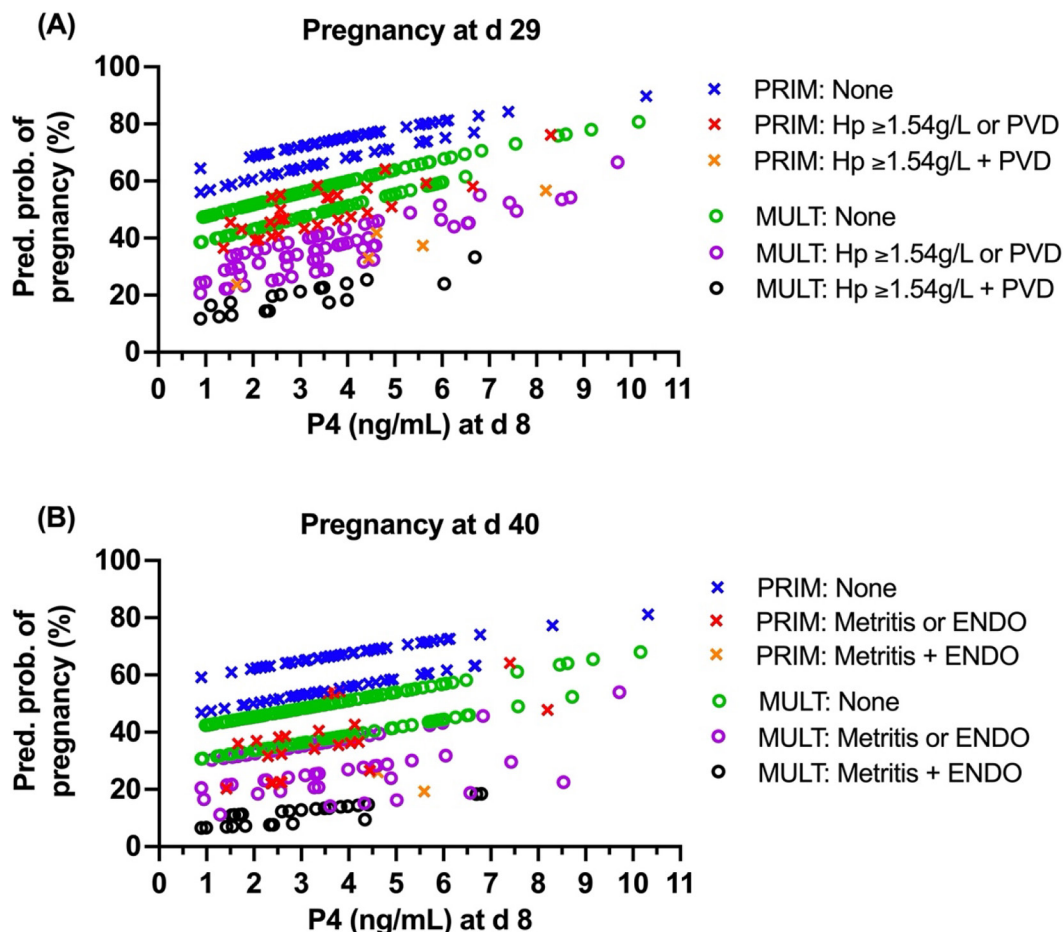


Figure 3. Associations between serum progesterone (P4) concentration measured at d 8 after first artificial insemination (AI) and the model-predicted probability of pregnancy on d 29 and 40 after AI in lactating Holstein cows with serum P4 ≥ 1.0 ng/mL on d 8 or 12 after AI from 2 commercial herds. In panel A ($n = 382$), the multivariable model accounted for parity [primiparous (PRIM) or multiparous (MULT)], haptoglobin (Hp) at 2 ± 2 DIM, and purulent vaginal discharge (PVD). In panel B ($n = 380$), the model accounted for parity, metritis, and endometritis (ENDO), and all models included herd as a random effect. Data are stratified by selected levels of covariates for clarity, and different patterns in datapoints within a category were due to levels not shown (e.g., only metritis, only ENDO, or herd). Each 1.0 ng/mL increase in serum P4 was associated with greater odds of P/AI on d 29 (AOR: 1.18; 1.04 to 1.34; $P = 0.01$) and 40 after AI (AOR: 1.12; 0.99 to 1.27; $P = 0.07$).

Table 7. Final logistic regression models of health variables associated with presumed pregnancy loss between d 19 and 29, 29 to 33, and 33 to 40 after first insemination in Holstein cows from two herds

Variable	n	LSM ¹	AOR ¹	95% CI ¹	P-value
From d 19 to 29²					
Metritis³					
Unaffected	217	21.7%	Reference	—	—
Affected	28	42.9%	2.7	1.20 to 6.17	0.02
Clinical disease⁴					
None	150	22.2%	Reference	—	—
Single	68	20.7%	0.92	0.39 to 2.13	0.97
Multiple	27	45.0%	3.13	0.99 to 10.0	0.05
From d 29 to 33²					
NEFA at 2 DIM⁵					
≤0.73 mM	199	4.5%	Reference	—	—
≥0.73 mM	52	12.8%	3.08	1.07 to 8.85	0.04
BCS at 63 DIM⁶					
≥3.00	219	4.9%	Reference	—	—
≤2.75	32	14.2%	3.23	1.01 to 10	0.05
From d 33 to 40²					
Metritis³					
Unaffected	217	7.2%	Reference	—	—
Affected	19	38.9%	8.26	2.65 to 25.64	<0.01
DA⁷					
Unaffected	231	8.9%	Reference	—	—
Affected	5	76.1%	33.33	4.33 to 250.0	<0.01
Clinical disease⁴					
None	150	8.4%	Reference	—	—
One	68	11.0%	1.35	0.44 to 4.13	0.81
Multiple	18	28.9%	4.44	1.03 to 19.23	0.05

¹Least squares means (LSM) of predicted probability of pregnancy loss, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

²Pregnancy at d 19 after first insemination was determined based on the expression of *ISG15* in blood leukocytes, and pregnancy at d 29 was based on serum pregnancy-associated glycoprotein.

³Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 ± 2 DIM).

⁴Clinical disease variable included any of retained placenta, milk fever, displaced abomasum, lameness, mastitis, metritis, or purulent vaginal discharge and was evaluated in a separate multivariable model.

⁵Nonesterified fatty acids (NEFA) measured in serum at 2 ± 2 DIM.

⁶Body condition score (BCS; 1 to 5 scale) assessed at 63 ± 3 DIM.

⁷DA = displaced abomasum diagnosed at surgical correction by 30 DIM.

Table 8. Final logistic regression models of health variables associated with presumed pregnancy loss¹ between d 19 and 40 after first insemination in 245 Holstein cows from two herds

Variable	n	LSM ²	AOR ²	95% CI ²	P-value
Metritis³					
Unaffected	217	33.3%	Reference	—	—
Affected	28	69.6%	4.59	1.94 to 10.99	<0.01
DA³					
Unaffected	239	37.4%	Reference	—	—
Affected	6	75.5%	5.15	0.89 to 29.41	0.07
Clinical disease⁴					
None	150	34.3%	Reference	—	—
One	68	33.7%	0.97	0.46 to 2.06	1.00
Multiple	27	66.8%	3.85	1.34 to 10.99	<0.01

¹Pregnancy at d 19 after first insemination was determined based on the expression of *ISG15* in blood leukocytes, and pregnancy at d 29 was based on serum pregnancy-associated glycoprotein.

²Least squares means (LSM) of predicted probability of pregnancy loss, adjusted odds ratio (AOR), and 95% confidence intervals (CI).

³Metritis = detection of fetid and colored (brownish/reddish) discharge at any of four examinations (4, 8, 11, and 15 ± 2 DIM); DA = displaced abomasum diagnosed at surgical correction by 30 DIM.

⁴Clinical disease variable included any of retained placenta, milk fever, displaced abomasum, lameness, mastitis, metritis, or purulent vaginal discharge and was evaluated in a separate multivariable model.

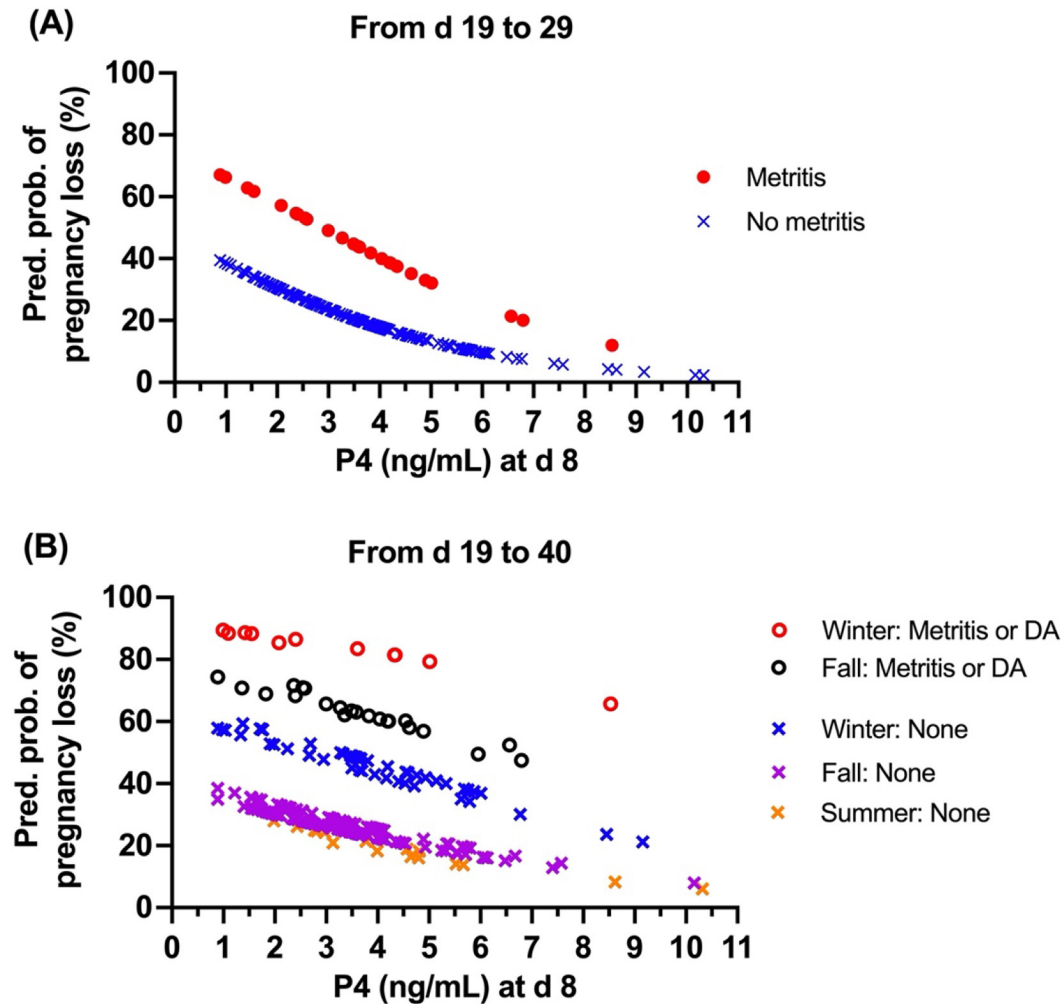


Figure 4. Associations between serum progesterone (P4) concentration measured at d 8 after first artificial insemination (AI) and the model-predicted probability of pregnancy loss from d 19 to 29 and from d 19 to 40 after AI on 230 lactating Holstein cows from 2 commercial herds. Pregnancy status at d 19 was estimated based on expression of *ISG15* in blood leukocytes and at d 29 and 40 was based on serum pregnancy-associated glycoprotein. In panel B, the multivariable model accounted for season at AI, metritis, and displaced abomasum (DA), and all models accounted for herd as a random effect. Data are stratified by selected levels of covariates for clarity, and different patterns in datapoints within a category were due to levels not shown (e.g., only metritis, only DA, or herd). Each 1 ng/mL increase in serum P4 was associated with decreased odds of pregnancy loss from d 19 to 29 (AOR: 0.70; 0.56 to 0.88; $P < 0.01$) and from d 19 to 40 after AI (AOR: 0.82; 0.68 to 0.99; $P = 0.04$).

number of cows with P-LOSS in each category is relatively small and these associations should be interpreted cautiously. From d 33 to 40, cows diagnosed with metritis, DA, or with >1 clinical disease were more likely to undergo P-LOSS. We speculate that such associations are attributable to impaired placental function that would result in P-LOSS before 40 to 45 d of gestation. After d 27, embryo adhesion is complete, placentomes start to develop, and embryo nutrition switches from uterine histotrophic to hemotrophic nutrition (Assis Neto et al., 2010). Therefore, inadequate placentation could affect nutrient supply and possibly increase the chances of P-LOSS at this stage. The concept of sub-optimal placental function could also be supported by

the negative association of levels of PAG at d 29 with P-LOSS from d 29 to 33 or from d 33 to 40 observed here. This is consistent with previous studies in which lower circulating pregnancy-specific protein B at d 24 or 25 of gestation was associated with P-LOSS by d 61 or 67 (Ribeiro et al., 2014; Stangaferro et al., 2021). However, associations of PAG levels and P-LOSS could be confounded by the delay of 7 to 10 d between P-LOSS and decrease in PAG (or PSPB) levels (Wijma et al., 2016). Using PAG as a marker, it is possible that placental function is altered in cows that had health disorders, which could be one of the pathways to early pregnancy losses.

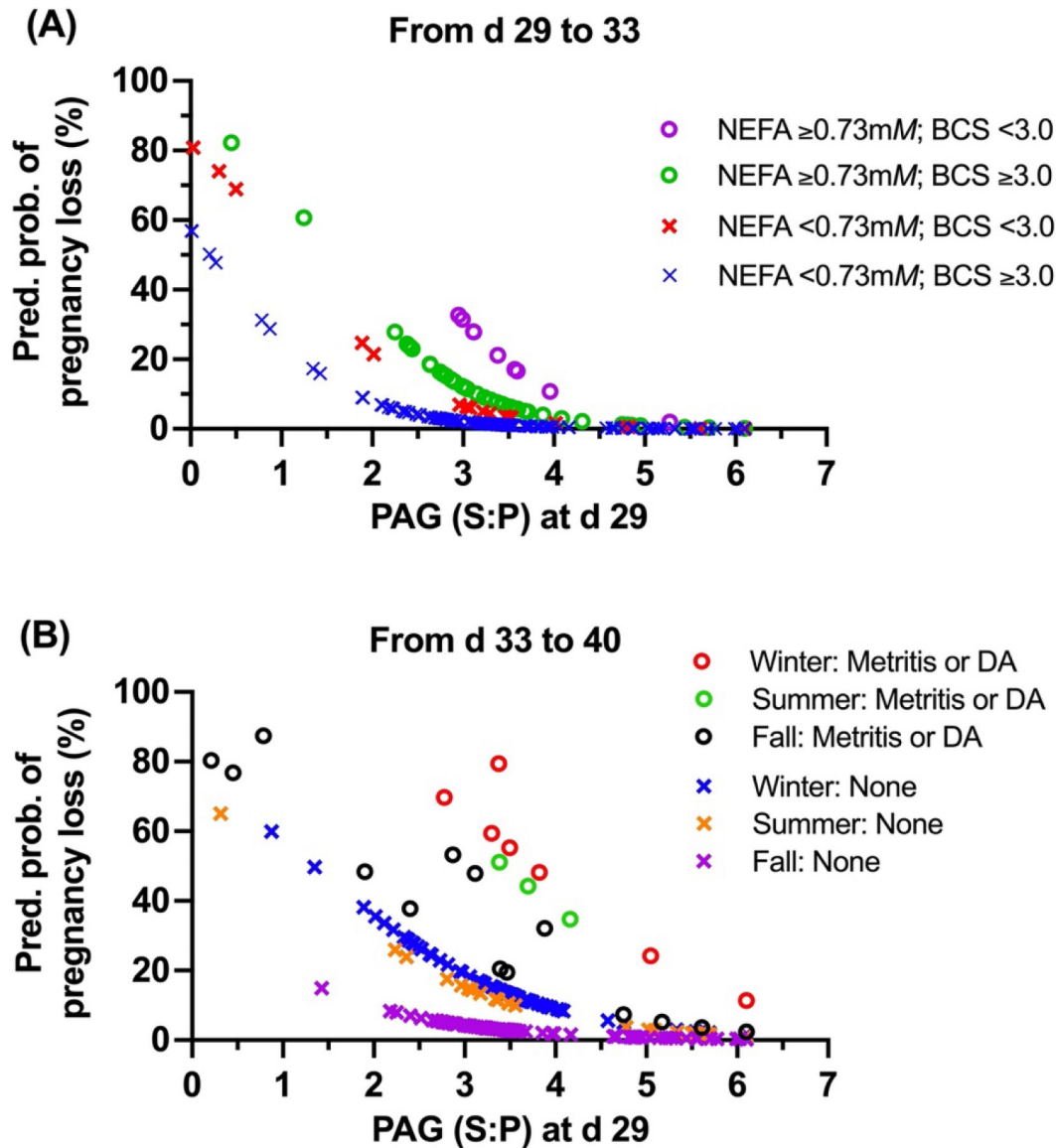


Figure 5. Associations between serum pregnancy-associated glycoprotein [PAG; measured as sample-to-positive ratio (S:P)] at d 29 after first artificial insemination (AI) and the model-predicted probability of pregnancy loss from d 29 to 33 and from d 33 to 40 after AI (panel A; $n = 248$) and from d 33 to 40 (panel B; $n = 233$) in lactating Holstein cows. Multivariable models accounted for serum NEFA concentrations at 2 ± 2 DIM and body condition score (BCS) at 63 DIM (panel A; $n = 248$), or season at AI, metritis, and DA (panel B; $n = 233$), and all models considered herd as a random effect. Data are stratified by selected levels of covariates for clarity. Each 1.0 unit increase in serum PAG S:P was associated with decreased odds of P-LOSS from d 29 to 33 (AOR: 0.25; 0.14 to 0.44; $P < 0.001$) and from d 33 to 40 after AI (AOR: 0.42; 0.27 to 0.65; $P < 0.01$).

Studies frequently evaluated risk factors for P-LOSS at later stages, based on pregnancy diagnosis by ultrasound at the first and second pregnancy examinations (30 and 60 d after AI). In a compilation of data on over 5,000 cows, Santos et al. (2010) reported 51% P/AI at first diagnosis in Holstein cows not diagnosed with a clinical disease, with 9% of them undergoing P-LOSS by d 60. However, P-LOSS was more likely in cows that had a calving problem (16%), PVD (15%), mastitis

(20%), or lameness (26%). In a study with grazing cows (Ribeiro et al., 2013), risk factors for reduced P/AI at d 30 were elevated NEFA (≥ 0.7 mM) at 7 or 14 DIM, digestive problems (diarrhea, bloat, or DA), metritis, or any clinical disease, while risk factors for P-LOSS from d 30 to 65 were calving problems (assisted calving, twin birth, stillbirth, or RP), PVD, or >1 clinical disease. Altogether, it seems clear that postpartum health dis-

orders are important risk factors for pregnancy losses from d 19 to 65 of gestation.

A limitation of our study is that the approach used to determine early pregnancy at d 19 after AI and P-LOSS from d 19 cannot account for potential false positives, where cows presumed pregnant by *ISG15* were truly non-pregnant. False positives would overestimate P/AI at d 19 and P-LOSS and result in a misclassification bias, likely increasing type II error and reducing our ability to find true associations. However, we observed biologically plausible associations of health disorders with P/AI at d 19 and P-LOSS from d 19 to 29 that would likely be stronger if false positives were minimized. A similar proportion of false negatives (12 to 19%) was observed previously for estimating pregnancy at d 19 based on *ISG15* (Han et al., 2006; Wiltbank et al., 2016). Furthermore, our sampling protocol for measurement of *ISG15* encompassing only d 19 $\pm$ 1 after AI (i.e., d 18, 19, or 20) could have misclassified cows with a later increase in circulating IFN- τ (and *ISG15* expression). This could have happened due to an expected variation in the timing of conceptus attachment between cows, as previously described with serial measurements of PSPB (Middleton and Pursley, 2019). The second limitation was the financial and logistical constraints on our sample size. Nevertheless, to our knowledge, our study is one of the first describing the associations of postpartum health with P-LOSS from 19 to 40 d after AI and will contribute to better understanding the consequences of health disorders for early pregnancy establishment and maintenance.

CONCLUSION

In summary, cows were less likely to be pregnant at d 19 after first AI if they had postpartum uterine disease, and less likely to be pregnant at d 29 through 40 after AI if they had elevated serum haptoglobin within 4 d postpartum, uterine disease, or multiple clinical diseases. Serum P4 concentrations during early diestrus were positively associated with P/AI from d 19 to 40. Similarly, the occurrence of a uterine disease or of multiple clinical diseases increased the odds of P-LOSS from d 19 to 29 and from d 33 to 40, and indicators of metabolic imbalance were associated with greater odds of P-LOSS from d 29 to 33. Furthermore, PAG levels in serum were positively associated with pregnancy maintenance. Our findings suggest that postpartum health disorders can have meaningful detrimental effects on early pregnancy establishment and maintenance.

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REFERENCES

- Albaaj, A., J. Durocher, S. J. LeBlanc, and S. Dufour. 2023. Meta-analysis of the incidence of pregnancy losses in dairy cows at different stages to 90 days of gestation. *JDS Commun.* 4:144–148. <https://doi.org/10.3168/jdsc.2022-0278>.
- Assis Neto, A. C., F. T. V. Pereira, T. C. Santos, C. E. Ambrosio, R. Leiser, and M. A. Miglino. 2010. Morpho-physical recording of bovine conceptus (*Bos indicus*) and placenta from Days 20 to 70 of pregnancy. *Reprod. Domest. Anim.* 45:760–772. <https://doi.org/10.1111/j.1439-0531.2009.01345.x>.
- Broes, A. J., and S. LeBlanc. 2014. Comparison of commercial progesterone assays for evaluation of luteal status in dairy cows. *Can. Vet. J.* 55:582–584.
- Bruin  , T. C., M. G. Colazo, E. S. Ribeiro, M. Gobikrushanth, and D. J. Ambrose. 2019. Using in-line milk progesterone data to characterize parameters of luteal activity and their association with fertility in Holstein cows. *J. Dairy Sci.* 102:780–798. <https://doi.org/10.3168/jds.2018-14654>.
- Bruin  , T.C., E.I. Morrison, E.S. Ribeiro, D.L. Renaud, R.C. Serr  nho, and S.J. LeBlanc. 2023. Postpartum health is associated with detection of estrus by activity monitors and reproductive performance in dairy cows. *J. Dairy Sci.* (Accepted Publ. Jul 2023).
- Chebel, R. C., J. E. P. Santos, J. P. Reynolds, R. L. A. Cerri, S. O. Juchem, and M. Overton. 2004. Factors affecting conception rate after artificial insemination and pregnancy loss in lactating dairy cows. *Anim. Reprod. Sci.* 84:239–255. <https://doi.org/10.1016/j.anireprosci.2003.12.012>.
- De Vries, A. 2006. Economic value of pregnancy in dairy cattle. *J. Dairy Sci.* 89:3876–3885. [https://doi.org/10.3168/jds.S0022-0302\(06\)72430-4](https://doi.org/10.3168/jds.S0022-0302(06)72430-4).
- Dubuc, J., T. F. Duffield, K. E. Leslie, J. S. Walton, and S. J. LeBlanc. 2011. Effects of postpartum uterine diseases on milk production and culling in dairy cows. *J. Dairy Sci.* 94:1339–1346. <https://doi.org/10.3168/jds.2010-3758>.
- Edmonson, A. J., I. J. Lean, L. D. Weaver, A. T. Farver, and G. Webster. 1989. A body condition scoring chart for Holstein Dairy Cows. *J. Dairy Sci.* 72:68–78. [https://doi.org/10.3168/jds.S0022-0302\(89\)79081-0](https://doi.org/10.3168/jds.S0022-0302(89)79081-0).
- Forde, N., M. E. Beltman, G. B. Duffy, J. P. Mehta, P. O’Gaora, J. F. Roche, P. Lonergan, and M. A. Crowe. 2011. Changes in the Endometrial Transcriptome During the Bovine Estrous Cycle: Effect of Low Circulating Progesterone and Consequences for Conceptus Elongation. *Biol. Reprod.* 84:266–278. <https://doi.org/10.1095/biolreprod.110.085910>.
- Forde, N., F. Carter, T. Fair, M. A. Crowe, A. C. O. Evans, T. E. Spencer, F. W. Bazer, R. McBride, M. P. Boland, P. O’Gaora, P. Lonergan, and J. F. Roche. 2009. Progesterone-Regulated Changes in Endometrial Gene Expression Contribute to Advanced Concep-

- tus Development in Cattle1. *Biol. Reprod.* 81:784–794. <https://doi.org/10.1095/biolreprod.108.074336>.
- Fricke, P. M., and M. C. Wiltbank. 2022. Symposium review: The implications of spontaneous versus synchronized ovulations on the reproductive performance of lactating dairy cows. *J. Dairy Sci.* doi:<https://doi.org/10.3168/jds.2021-21431>.
- Gifford, C. A., K. Racicot, D. S. Clark, K. J. Austin, T. R. Hansen, M. C. Lucy, C. J. Davies, and T. L. Ott. 2007. Regulation of Interferon-Stimulated Genes in Peripheral Blood Leukocytes in Pregnant and Bred, Nonpregnant Dairy Cows. *J. Dairy Sci.* 90:274–280. [https://doi.org/10.3168/jds.S0022-0302\(07\)72628-0](https://doi.org/10.3168/jds.S0022-0302(07)72628-0).
- Han, H., K. J. Austin, L. A. Rempel, and T. R. Hansen. 2006. Low blood ISG15 mRNA and progesterone levels are predictive of non-pregnant dairy cows. *J. Endocrinol.* 191:505–512. <https://doi.org/10.1677/joe.1.07015>.
- LeBlanc, S. J. 2012. Interactions of Metabolism, Inflammation, and Reproductive Tract Health in the Postpartum Period in Dairy Cattle. *Reprod. Domest. Anim.* 47(s5):18–30. <https://doi.org/10.1111/j.1439-0531.2012.02109.x>.
- Makimura, S., and N. Suzuki. 1982. Quantitative determination of bovine serum haptoglobin and its elevation in some inflammatory diseases. *Nippon Juigaku Zasshi* 44:15–21. <https://doi.org/10.1292/jvms1939.44.15>.
- Middleton, E. L., and J. R. Pursley. 2019. Short communication: Blood samples before and after embryonic attachment accurately determine non-pregnant lactating dairy cows at 24 d post-artificial insemination using a commercially available assay for pregnancy-specific protein B. *J. Dairy Sci.* 102:7570–7575. <https://doi.org/10.3168/jds.2018-15961>.
- Molina-Coto, R., S. G. Moore, L. M. Mayo, W. R. Lamberson, S. E. Pooock, and M. C. Lucy. 2020. Ovarian function and the establishment and maintenance of pregnancy in dairy cows with and without evidence of postpartum uterine disease. *J. Dairy Sci.* 103:10715–10727. <https://doi.org/10.3168/jds.2020-18694>.
- O'Connor, A. M., J. M. Sargeant, I. R. Dohoo, H. N. Erb, M. Cevallos, M. Egger, A. K. Ersbøll, S. W. Martin, L. R. Nielsen, D. L. Pearl, D. U. Pfeiffer, J. Sanchez, M. E. Torrence, H. Vigre, C. Waldner, and M. P. Ward. 2016. Explanation and Elaboration Document for the STROBE-Vet Statement: Strengthening the Reporting of Observational Studies in Epidemiology—Veterinary Extension. *J. Vet. Intern. Med.* 30:1896–1928. <https://doi.org/10.1111/jvim.14592>.
- Pinedo, P., J. E. P. Santos, R. C. Chebel, K. N. Galvão, G. M. Schuenemann, R. C. Bicalho, R. O. Gilbert, S. Rodriguez Zas, C. M. Seabury, G. Rosa, and W. W. Thatcher. 2020. Early-lactation diseases and fertility in 2 seasons of calving across US dairy herds. *J. Dairy Sci.* 103:10560–10576. <https://doi.org/10.3168/jds.2019-17951>.
- Ribeiro, E. S., R. G. S. Bruno, A. M. Farias, J. A. Hernández-Rivera, G. C. Gomes, R. Surjus, L. F. V. Becker, A. Birt, T. L. Ott, J. R. Brannen, R. G. Sasser, D. H. Keisler, W. W. Thatcher, T. R. Bilby, and J. E. P. Santos. 2014. Low doses of bovine somatotropin enhance conceptus development and fertility in lactating dairy cows. *Biol. Reprod.* 90:1–12. <https://doi.org/10.1095/biolreprod.113.114694>.
- Ribeiro, E. S., K. N. Galvão, W. W. Thatcher, and J. E. P. Santos. 2012. Economic aspects of applying reproductive technologies to dairy herds. *Anim. Reprod.* 9:370–387.
- Ribeiro, E. S., G. Gomes, L. F. Greco, R. L. A. Cerri, A. Vieira-Neto, P. L. J. Monteiro Jr., F. S. Lima, R. S. Bisinotto, W. W. Thatcher, and J. E. P. Santos. 2016. Carryover effect of postpartum inflammatory diseases on developmental biology and fertility in lactating dairy cows. *J. Dairy Sci.* 99:2201–2220. <https://doi.org/10.3168/jds.2015-10337>.
- Ribeiro, E. S., F. S. Lima, L. F. Greco, R. S. Bisinotto, A. P. A. Monteiro, M. Favoreto, H. Ayres, R. S. Marsola, N. Martinez, W. W. Thatcher, and J. E. P. Santos. 2013. Prevalence of periparturient diseases and effects on fertility of seasonally calving grazing dairy cows supplemented with concentrates. *J. Dairy Sci.* 96:5682–5697. <https://doi.org/10.3168/jds.2012-6335>.
- Ricci, a, P.D. Carvalho, M.C. Amundson, R.H. Fourdraine, L. Vincenti, and P.M. Fricke. 2015. Factors associated with pregnancy-associated glycoprotein (PAG) levels in plasma and milk of Holstein cows during early pregnancy and their effect on the accuracy of pregnancy diagnosis. *J. Dairy Sci.* 98:2502–14. doi:<https://doi.org/10.3168/jds.2014-8974>.
- Santos, J. E., R. S. Bisinotto, E. S. Ribeiro, F. S. Lima, L. F. Greco, C. R. Staples, and W. W. Thatcher. 2010. Applying nutrition and physiology to improve reproduction in dairy cattle. *Soc. Reprod. Fertil. Suppl.* 67:387–403. <https://doi.org/10.5661/RDR-VII-387>.
- Sellmer Ramos, I., J. G. N. Moraes, M. O. Caldeira, S. E. Pooock, T. E. Spencer, and M. C. Lucy. 2023. Impact of postpartum metritis on the regeneration of endometrial glands in dairy cows Impact of postpartum metritis on the regeneration of endometrial glands in dairy cows. *JDSC*. doi:<https://doi.org/10.3168/jdsc.2022-0338>.
- Sheldon, I.M., G.S. Lewis, S. LeBlanc, and R.O. Gilbert. 2006. Defining postpartum uterine disease in cattle. 65:1516–1530. doi:<https://doi.org/10.1016/j.theriogenology.2005.08.021>.
- Sinedino, L. D. P., F. S. Lima, R. S. Bisinotto, R. L. A. Cerri, and J. E. P. Santos. 2014. Effect of early or late resynchronization based on different methods of pregnancy diagnosis on reproductive performance of dairy cows. *J. Dairy Sci.* 97:4932–4941. <https://doi.org/10.3168/jds.2013-7887>.
- Skinner, G., R. A. L. Brown, and L. Roberts. 1991. Bovine haptoglobin response in clinically defined field conditions. *Vet. Rec.* 128:147–149. <https://doi.org/10.1136/vr.128.7.147>.
- Spencer, T. E., and F. W. Bazer. 2004. Conceptus signals for establishing and maintenance of pregnancy. *Reprod. Biol. Endocrinol.* 2:1–15. <https://doi.org/10.1186/1477-7827-2-1>.
- Sprecher, D. J., D. E. Hostetler, and J. B. Kaneene. 1997. A lameness scoring system that uses posture and gait to predict dairy cattle reproductive performance. *Theriogenology* 47:1179–1187. [https://doi.org/10.1016/S0093-691X\(97\)00098-8](https://doi.org/10.1016/S0093-691X(97)00098-8).
- Stangaferro, M. L., M. Z. Toledo, R. S. Gennari, M. M. Perez, R. V. Barletta, R. Wijma, E. M. Sitko, G. Granados, M. Masello, M. E. Van Amburgh, D. Luchini, J. O. Giordano, R. D. Shaver, and M. C. Wiltbank. 2021. Effects of feeding rumen-protected methionine pre- and postpartum in multiparous Holstein cows: Lactation performance and plasma amino acid concentrations. *J. Dairy Sci.* 104:7583–7603. <https://doi.org/10.3168/jds.2021-20190>.
- Vandesompele, J., K. De Preter, F. Pattyn, B. Poppe, N. Van Roy, A. De Paepe, and F. Speleman. 2002. Accurate normalization of real-time quantitative RT-PCR data by geometric averaging of multiple internal control genes. *Genome Biol.* 3:1–12. <https://doi.org/10.1186/gb-2002-3-7-research0034>.
- Wijma, R., M. L. Stangaferro, M. M. Kamat, S. Vasudevan, T. L. Ott, and J. O. Giordano. 2016. Embryo Mortality Around the Period of Maintenance of the Corpus Luteum Causes Alterations to the Ovarian Function of Lactating Dairy Cows. *Biol. Reprod.* 95:112. <https://doi.org/10.1095/biolreprod.116.142075>.
- Wiltbank, M. C., G. M. Baez, A. Garcia-Guerra, M. Z. Toledo, P. L. J. Monteiro, L. F. Melo, J. C. Ochoa, J. E. P. Santos, and R. Sartori. 2016. Pivotal periods for pregnancy loss during the first trimester of gestation in lactating dairy cows. *Theriogenology* 86:239–253. <https://doi.org/10.1016/j.theriogenology.2016.04.037>.

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